Practical No: 4-A

Advanced Layout and View Handling

Aim: Develop an Android application using ScrollView to display long scrollable content.

ScrollView is a layout container that allows users to scroll vertically when the content exceeds the screen size.

Features:

- Enables vertical scrolling.
- Contains only one direct child view.
- Commonly used with LinearLayout.
- Useful for forms, articles, and long text content.

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<ScrollView xmlns:android="http://schemas.android.com/apk/res/android"
  xmlns:app="http://schemas.android.com/apk/res-auto"
  xmlns:tools="http://schemas.android.com/tools"
  android:id="@+id/main"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  tools:context=".MainActivity">

  <LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="vertical"
    android:padding="16dp"
    >

    <TextView
      android:layout_width="wrap_content"
      android:layout_height="wrap_content"
      android:text="ANDROID PROGRAMMING"
      android:textSize="24sp"
      android:textStyle="bold"
      android:paddingTop="20dp"/>

    <TextView
      android:layout_width="wrap_content"
```

```
    android:layout_height="wrap_content"
    android:text="Unit 1: Introduction to Android"
    android:paddingTop="20dp"/>

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Unit 2: Activities and Intents"
    android:paddingTop="20dp"/>

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Unit 3: Layouts and Views"
    android:paddingTop="20dp"/>

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Unit 4: Event Handling"
    android:paddingTop="20dp"/>

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Unit 5: Fragments"
    android:paddingTop="20dp"/>

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Unit 6: Menus and Dialogs"
    android:paddingTop="20dp"/>

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Unit 7: SQLite Database"
    android:paddingTop="20dp"/>

<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Unit 8: Networking"
```

```
        android:paddingTop="20dp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Unit 9: Firebase"
        android:paddingTop="20dp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Unit 10: Project Development"
        android:paddingTop="20dp"/>

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="This content is intentionally long to demonstrate vertical scrolling in Android using ScrollView. Students can continue scrolling to view additional information on the screen."
        android:paddingTop="20dp"/>

  </LinearLayout>

</ScrollView>
```

Output:

